

Name: _____ Teacher: _____

Sydney Technical High School
Year 12 Ext.2 Mathematics HSC Assessment Task 1 March 2004

Instructions:

Start each question on a new page.

Show all necessary working. Single column of work only.

Staple these questions to the front of your answers.

Full marks may not be awarded for careless* or incomplete work.

Indicated marks are a guide and may change slightly during the marking process.

* Be careful when writing “z” so that is distinguishable from “2”.

Time allowed: 70 mins

Q1	Q2	Q3	TOTAL
/14	/17	/16	/47

Question 1

- 3 a) Given that a and b are real numbers, find a and b if

$$\frac{3+4i}{a+bi} = 1+i$$

- 8 b) If $z = -1 + \sqrt{3}i$ and $w = 2\left[\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right]$

i) Find $|z|$

ii) Write z in mod-arg form

iii) Evaluate the following in simplest mod-arg form

$\alpha)$ zw

$\beta)$ $\frac{z}{w}$

$\gamma)$ w^7

iv) Show w and $\sqrt{w}$ on a number plane diagram and on it write the values of $\sqrt{w}$ in mod-arg form.

- 3 c) For a complex number z , $\text{Arg}(z+2) = \frac{1}{2} \text{Arg}(z)$.

i) Find, giving reasons, the value of $|z|$.

ii) Give an expression for $\text{Arg}(z-2)$ in terms of $\text{Arg}(z)$.

Question 2 (Begin a new page)

- 8 a) For the ellipse $4x^2 + 9y^2 = 36$
- i) find the co-ordinates of the foci
 - ii) find the equations of the directrices
 - iii) Sketch the ellipse showing x & y intercepts, foci and directrices.
 - iv) Sketch the following ellipses, explaining the relationship to $4x^2 + 9y^2 = 36$ for each one.

$\alpha) \quad 9x^2 + 4y^2 = 36$

$\beta) \quad c^2x^2 + 9y^2 = 36 \text{ where } c^2 > 4$

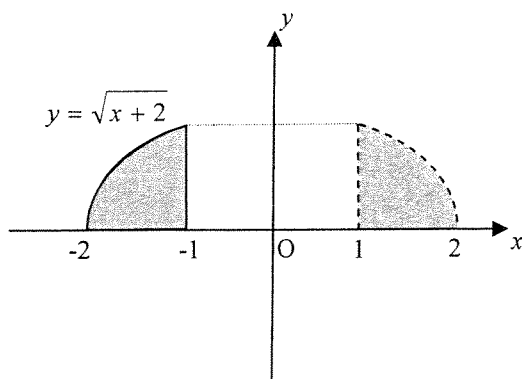
- 9 b) i) Differentiate $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ implicitly to show that $\frac{dy}{dx} = -\frac{b^2x}{a^2y}$.
- ii) Hence prove that the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(x_1, y_1)$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
- iii) The tangent at P cuts the directrix in the first quadrant at D. Find the co-ordinates of D.
- iv) If S is the focus associated with the directrix in iii), prove that $\angle PSD = 90^\circ$

Question 3 (Begin a new page)

- 4 a) Sketch the following loci
- i) $|z - 2| = 2, 0 \leq \arg z \leq \frac{\pi}{2}$
 - ii) $\arg(z + 1) = \pi/4, \operatorname{Re}(z) \leq 2$
- 3 b) i) The locus of the point P (x, y) which represents the complex number z is given by the equation $\operatorname{Im}(z) = |z - 2i|$. Find the Cartesian equation and sketch the locus of P.
- ii) Find the least value of $\arg z$ in part b (i)

- 5 c) i) Show on an Argand diagram the positions of the roots of $z^3 = -1$.
 ii) Explain algebraically why the roots of $z^3 = -1$ are among the roots of $z^6 = 1$.
 iii) By referring to the roots of $z^6 = 1$, find the roots of $z^4 + z^2 + 1 = 0$ in mod-arg form.

4 d)



The area under the curve $y = \sqrt{x+2}$ between $x = -2$ and $x = -1$ is rotated about the y axis to form a kind of “donut”. Find the volume of the donut in terms of π .

End of Examination

SUGGESTED SOLUTIONS AND MARKING SCHEME

EXT. 2 EXAM MARCH 2004.

Q1 a) $\frac{3+4i}{a+bi} = 1+i$

$$3+4i = (a+bi)(1+i)$$

$$= a + (a+b)i - b$$

$$\therefore \begin{cases} a+b=4 \\ a-b=3 \end{cases} \quad (1)$$

$$\therefore 2a = 7$$

$$\begin{cases} a = \frac{7}{2} \\ \therefore b = \frac{1}{2} \end{cases} \quad (2)$$

Q1 b) i) $|z| = 2 \quad (1)$

ii) $z = 2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) \quad (1)$

iii) $zw = 4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) \quad (1)$

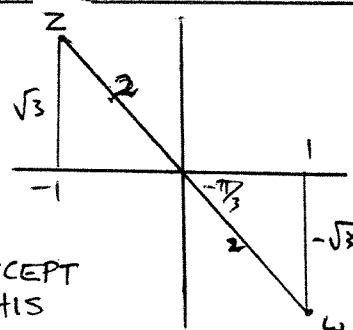
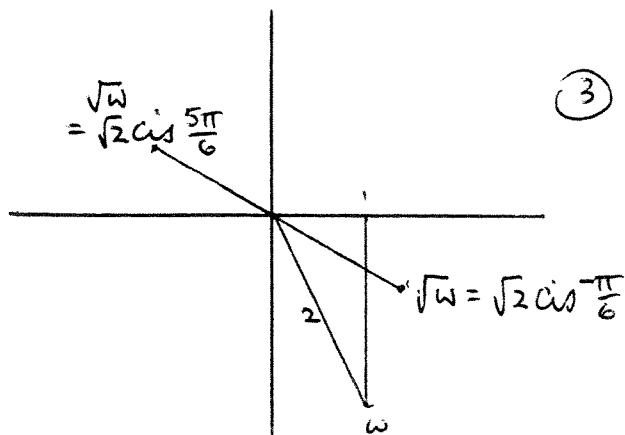
$\beta) \frac{z}{w} = 1(\cos \pi + i \sin \pi)$

$$= -1 \quad (1)$$

$\gamma) w^7 = 2^7(\cos \frac{-7\pi}{3} + i \sin \frac{-7\pi}{3})$

$$= 128(\cos \frac{-\pi}{3} + i \sin \frac{-\pi}{3}) \quad (1)$$

iv)

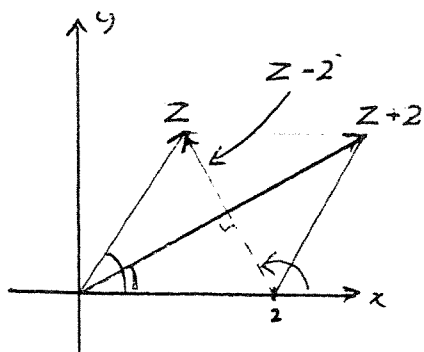


← ACCEPT THIS

MUST BE IN SIMPLEST FORM

1 for $\sqrt{w}$ approx bisecting $\arg w$. 1 each for correct values of $\sqrt{w}$

Q1 c)



i) $|z| = 2$, vectors form rhombus since $\arg z$ is bisected. (2)

ii) $\frac{\pi}{2} + \frac{1}{2} \arg(z)$. (1)

1 for value

1 for reason

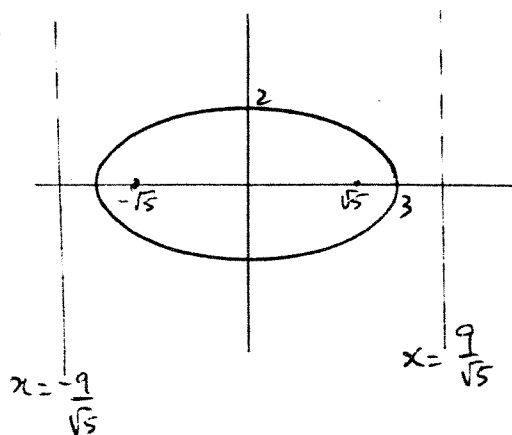
1 - no reason reqd.

Q2(a) $4x^2 + 9y^2 = 36$

i) foci $(\pm\sqrt{5}, 0)$ (1)

ii) $x = \pm \frac{9}{\sqrt{5}}$ (1)

iii)

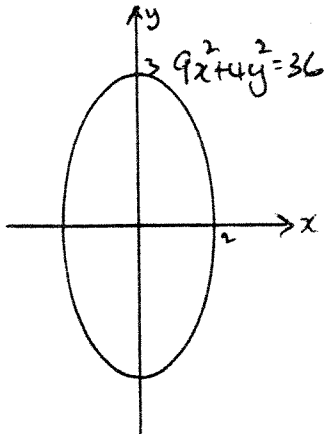


(1)

Be lenient. eg OK if they forget $\pm$ etc.

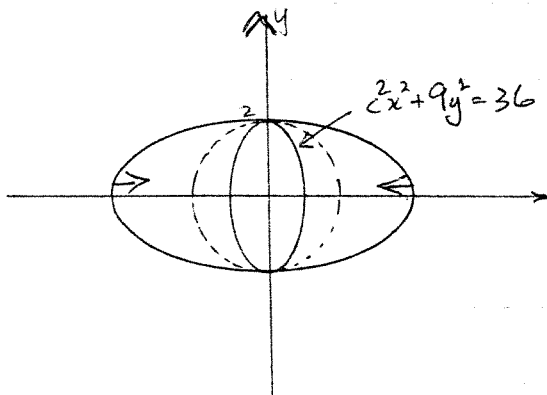
We're looking for correct orientation here / or allow (1) if the curve matches the data from i) & ii) (even if incorrect)

2(a) (iv) x)
Cont



$4x^2 + 9y^2 = 36$ is rotated 90° about centre etc

β)



Major axis shortens until
(ellipse becomes the circle $x^2 + y^2 = 4$ when $c^2 = 9$
then) orientation changes
and y axis becomes the
major axis

①

①

①

① for correct idea of
"squashing" ellipse
towards y axis

① for specific mention of
circle when $c^2 = 9$

OR

specific mention of
reorientation so that
major axis/foci now lie
on y axis.

Q2 b) i) $\frac{d}{dx} \frac{x^2}{a^2} + \frac{d}{dx} \frac{y^2}{b^2} = \frac{d}{dx} 1$

ie $2x + 2y \frac{dy}{dx} = 0$

← ① for this step.

$\therefore \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$

ii) at $P(x_1, y_1)$ slope of tangent is $-\frac{b^2 x_1}{a^2 y_1}$

①

$\therefore$ Eqn of tangent is

$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$

①

↓

$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$

But $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$

} ← ① for recognising this

$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

c) at D, $x = \frac{a}{e}$

$\therefore \frac{x_1}{ea} + \frac{yy_1}{b^2} = 1$

↓

$y = \frac{b^2}{y_1} \left(1 - \frac{x_1}{ae}\right)$

①

$\therefore D \left(\frac{a}{e}, \frac{b^2}{y_1} \left(1 - \frac{x_1}{ae}\right) \right)$

Q2 cont. b(iv) Now slope_{PS} = $\frac{y_1}{x_1 - ae}$

$$\text{slope}_{DS} = \frac{\frac{b^2}{y_1} \left(1 - \frac{x_1}{ae}\right)}{\frac{a}{e} - ae}$$

And $\left(\frac{y_1}{x_1 - ae}\right) \times \left(\frac{\frac{b^2}{y_1} \left(1 - \frac{x_1}{ae}\right)}{\frac{a}{e} - ae}\right)$

$$= \frac{-b^2/ae}{a\left(\frac{1}{e} - e\right)}$$

$$= \frac{-b^2}{a^2(1-e^2)}$$

$$= -\frac{b^2}{b^2} \quad \text{①}$$

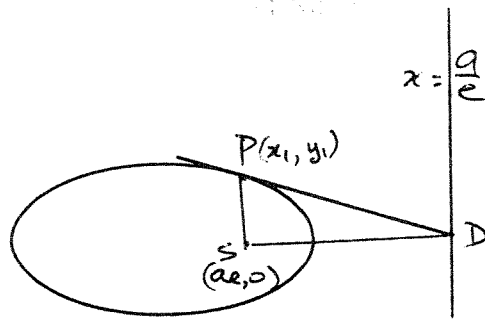
$$= -1$$

$$\therefore \angle DSP = 90^\circ$$

①

①

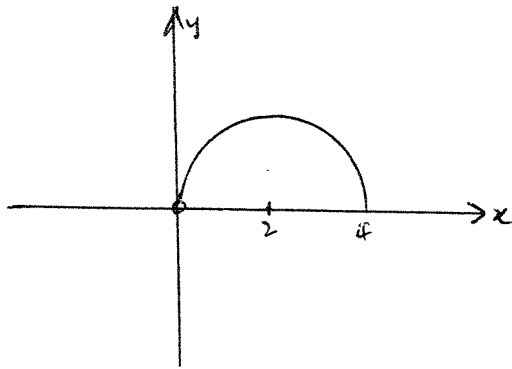
①



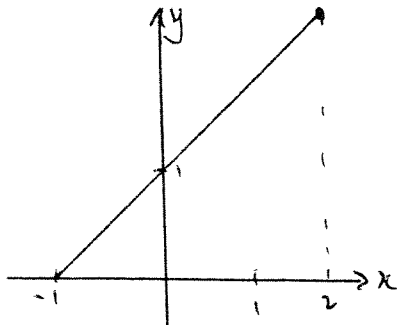
2 marks for working.

QUESTION 3.

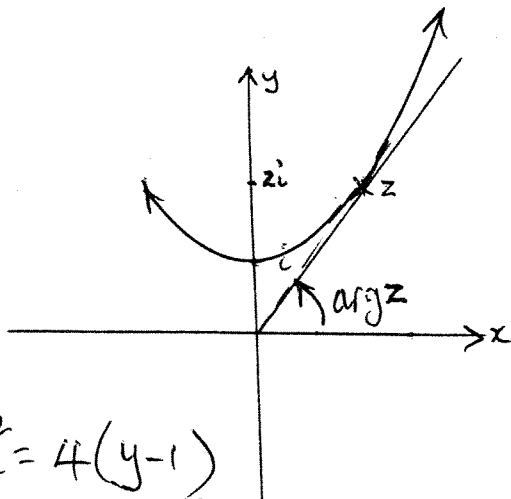
23(a) i)



ii)



3(b)



$$x^2 = 4(y-1)$$

$$\text{or } y = \frac{x^2}{4} + 1$$

$$\text{Min arg } z = \pi/4$$

① for circle at $x=2$

① for top half.

No penalty for (0,0) if included.

① For line from $x=-1$.

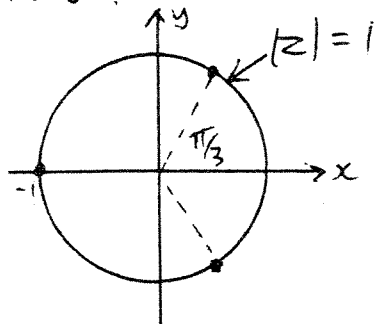
① For truncating at $x=2$.

① for equation

① for $\arg z = \pi/4$

Question 3.

Q3(c) i)



ii) Since $z^6 = 1$ can be factorized $(z^3+1)(z^3-1) = 0$ some of the roots of $z^6 = 1$ are given by $z^3+1 = 0$ which are the roots of $z^3 = -1$ as well.

iii) Since $z^6 - 1 = 0$ can be factorized as

$$(z^2)^3 - 1 = 0$$

$$\text{i.e. } (z^2-1)(z^4+z^2+1) = 0$$

when $z \neq \pm 1$, the roots of $z^4+z^2+1=0$ are the 4 complex roots of $z^6 = 1$

$$\text{i.e. } \text{cis } \pm \frac{\pi}{3}, \text{cis } \pm \frac{2\pi}{3}$$

① for roots of $z^3 = -1$

①

①

①

①

2 marks
- reference to $z^6 = 1$ must be made.

d) $V = \pi \int_0^1 x^2 dy - \pi \cdot 1^2 \cdot 1$

$$= \pi \int_0^1 (y^2 - 2)^2 dy - \pi$$

$$= \pi \int_0^1 y^4 - 4y^2 + 4 dy - \pi$$

$$= \frac{28\pi}{15}$$

①

①

①

①